Control: vabs(-4) send back 4.

4. Back · Flash [NUM] Write in decimal notation 2.5 $\times 10^3$ then 6.2 $\times 10^{-2}$.

Rep. 2 500 ; 0,062.

5. Red thread · Flash [REP] Give the symmetrical of $A(2;-3)$ in relation to origin.

Rep. $(-2;3)$.

Form 79 game. 10 June 2027 Presentation of specialized teachings; stocktaking

1. Warming · Flash [NUM] Express $\frac{3}{8}$ in percentage.

Rep. 37,5 %.

2. Heart · Metacognition [MET] What are your three revision priorities before the First?

Rep. Personal response: identify in the year sheets the questions that were most hesitant.

3. Heart + · Flash [EQ] Solve 2 $x^2 = 18$.

Rep. $x^2 = 9 : x = 3$ or $x = -3$.

4. Back · Trap [Sighs] Calculate $(-2)^4$ and -2^4.

Rep. 16 and -16.

The parentheses decide to whom the exhibitor applies.

5. Red thread · True/false [IVD] True or false? Justify. If an integer is divisible by 4 and by 6, then it is divisible by 24.

Rep. False. Counter-example: 12 is divisible by 4 and by 6, not by 24; it is divisible by PPCM $(4\,;\,6) = 12$.

Annual review and consolidation of the Second

Form 80 Mon. 14 June 2027 Annual review and consolidation

1. Warming · Flash [GEO] $A(-1;4)$ and $B(3;1)$. Contact details $\overrightarrow{AB}$?

Rep. $\begin{pmatrix} 4 \\ -3 \end{pmatrix}$.

2. Heart · Method [GEO] Equation reduced (AB) with $A(0;1)$ and $B(2;5)$.

Rep. Pent 2 ; $y = 2x + 1$.

3. Heart + · Method [REP] $A(1;0)$, $B(4;1)$, $C(5;4)$. Determine D so that $ABCD$ or a parallelogram.

Rep. $D = A + C - B = (2;3)$; control: $\overrightarrow{AB} = \overrightarrow{DC} = \begin{pmatrix} 3 \\ 1 \end{pmatrix}$.

4. Back · Flash [CAL] Calculate $-5 + 8 \times (-2)$.

Rep. $-5 - 16 = -21$.

Priority of multiplication.

5. Red thread · Statement→ formula [REG] The area of a side square x is a function A of x. Provide $A(x)$, table of values for $x = 1\,;\,2\,;\,3$, and whether A is proportional to x.

Rep. $A(x) = x^2$; 1 ; 4 ; 9 ; not proportional : $A(2) \neq 2A(1)$.

Form 81 Mar. 15 June 2027 Annual review and consolidation

1. Warming · Flash [NUM] Write 0.00045 in scientific notation.

Rep. $4{,}5 \times 10^{-4}$.

2. Heart · Flash [NUM] Determine $[-2;3[\cap]0;5]$.

Rep. $]0;3[$.

3. Heart + · Method [EQ] Solve $|x-1| < 4$ and give S in the form of interval.

Rep. $-4 < x - 1 < 4 : S =]-3;5[$.

4. Back · Develop [DEV] Develop and reduce $(2x-1)^2$.

Rep. $4x^2 - 4x + 1$.

5. Red thread · Flash [GEO] A cylinder has for radius 3 cm and height 5 cm. Calculate its exact volume.

Rep. $V = \pi \times 3^2 \times 5 = 45\pi$ cm^3.

Form 82 17 June 2027 Annual review and consolidation

1. Warming · Flash [IMG] $f(x) = x^2 - 4x$. Calculate $f(3)$.

Rep. $9 - 12 = -3$.

2. Heart · Flash [EQ] Solve $x^2 = 9$ then $|x| = 2$.

Rep. $x = \pm 3$; $x = \pm 2$.

3. Heart + · Method [REG] $f(x) = 2x - 3$ and $g(x) = x + 1$. Resolve $f(x) < g(x)$.

Rep. $2x - 3 < x + 1 \Leftrightarrow x < 4 : S =]-\infty; 4[$.

4. Back · Flash [QP] A tap flows 18 L in 4 min. What volume in 10 min?

Rep. $4{,}5 \times 10 = 45$ L.

5. Red thread · Code reading [ALG] What is the final value of `n` ?

```
n = 0
while n*n < 30:
    n = n + 1
```

Rep. $n = 6$ because $5^2 = 25 < 30$ but $6^2 = 36 \geqslant 30$.

Smaller integer whose square exceeds 30.

Form 83 Mon. 21 June 2027 Annual review and consolidation

1. Warming · Flash [STA] Average of 5, 7, 9, 11?

Rep. 8.

2. Heart · Method [STA] Series 1 ; 3 ; 4 ; 6 ; 8 ; 9 ; 11 : Q_1, median Q_3.

Rep. $n = 7$: Q_1 row $\lceil 7/4 \rceil = 2$: 3 ; median (range 4) : 6 ; Q_3 row $\lceil 21/4 \rceil = 6$: 9.

3. Heart + · True/false [STA] True or false? Justify. Add 10 to each value of a series leaves its gap type unchanged.

Rep. True, average deviations do not change.

4. Back · Reflection [IMG] How many years have -1 and 4 for $f(x) = x^2$?

Rep. -1 : none; 4 : two (-2 and 2).

5. Red thread · Prove [LIT] Show that for all real x : $(x-1)(x+1) + 1 = x^2$.

Rep. $(x-1)(x+1) + 1 = x^2 - 1 + 1 = x^2$.

Form 84 March 22 June 2027 Annual review and consolidation

1. Warming · Flash [DEV] Develop and reduce $(2x - 3)(x + 4)$.

Rep. $2x^2 + 8x - 3x - 12 = 2x^2 + 5x - 12$.

2. Heart · Flash [EQ] Solve 3 $x - 7 \leqslant 2x + 5$.

Rep. $x \leqslant 12$; $S =]-\infty; 12]$.

3. Heart + · Method [EQ] Solve $(x-1)(x+4) = 0$ then $x^2 = 5$.

Rep. $x = 1$ or $x = -4$; $x = \sqrt{5}$ or $x = -\sqrt{5}$.

4. Back · Reflection [PNP] Subscription : 5 € + 3 € per entry. Is the price proportional to the What's going on?

Rep. No: 1 entry $\mapsto$ 8 €, 2 entries $\mapsto$ 11 $\neq$ 16 €.

5. Red thread · Flash [SOL] A cone has for radius 3 cm and height 4 cm. Calculate its generator and sound exact volume.

Rep. Generator $\sqrt{3^2 + 4^2} = 5$ cm; $V = \dfrac{1}{3}\pi \times 9 \times 4 = 12\pi$ cm^3.

Form 85 24 June 2027 Annual review and consolidation

1. Warming · Flash [PRO] What multiplier coefficient corresponds to an increase of 8%?

Rep. 1,08.

2. Heart · Method [PRO] A 25% increase is followed by a 20% decrease. What is the overall rate?

Rep. $1{,}25 \times 0{,}8 = 1$: no change (0%).

3. Heart + · Flash [PRO] A price goes from 5,000 to 5,750 Ar. Rate of change?

Rep. $\dfrac{750}{5\,000} = 15\,\%$.

4. Back · Recognize [Sighs] What is the section of a cylinder by a plane containing its axis? Plane perpendicular to its axis?

Rep. A rectangle; a disc.

5. Red thread · Problem [EQ] The triple of a number, minus 4, is equal to twice that number increased 5. Write an equation and find the number.

Rep. $3x - 4 = 2x + 5$, where $x = 9$.
Verification: $3 \times 9 - 4 = 23 = 2 \times 9 + 5$.

Form 86 Mon. 28 June 2027 Annual review and consolidation

1. Warming · Flash [MOD] Meaning of variation of $x \mapsto -3x + 2$?

Rep. Strictly decreasing (slope $-3 < 0$).

2. Heart · Method [REG] Minimum and maximum of $f(x) = x^2$ on $[-3;1]$?

Rep. Minimum 0 (in 0); maximum 9 (in -3) ; $f(1) = 1 < 9$.

3. Heart + · Method [CMP] For $x > 0$, compare $\dfrac{1}{x}$ and $\dfrac{1}{x+1}$.

Rep. $0 < x < x + 1$ and the inverse is decreasing on $]0;+\infty[$: $\dfrac{1}{x} > \dfrac{1}{x+1}$.

4. Back · Construction programme [chuckles] Describe in three stages the construction of the symmetrical A' of A by ratio to one right (d).

Rep. 1. Perpendicular to (d) passing through A, cutting (d) in H. 2. Report to compass HA on the other side of (d). 3. A' is the point obtained: H is the midpoint of $[AA']$.

5. Red thread · Method [DEM] One point I checks $IA = IB$. Where is it located? What property?

Rep. On the mediator of $[AB]$ (characterization of the mediator).

Form 87 March 29 June 2027 Annual review and consolidation

1. Warming · Flash [PRB] $P(A) = 0.18$. Calculate $P(\overline{A})$.

Rep. 0,72.

2. Heart · Method [PRB] 100 people including 60 women; 6 women and 4 men are left-handed. $P_F(G)$ and $P_G(F)$.

Rep. $P_F(G) = \dfrac{6}{60} = 0{,}1$; $P_G(F) = \dfrac{6}{10} = 0{,}6$.

3. Heart + · Method [PRB] $P(A) = 0{,}4$, $P_A(B) = 0{,}5$, $P_{\overline{A}}(B) = 0.2$. Calculate $P(B)$.

Rep. $0{,}4 \times 0{,}5 + 0{,}6 \times 0{,}2 = 0{,}2 + 0{,}12 = 0{,}32$.

4. Back · Problem [MOD] Volume of a edge cube x : $V(x) = x^3$. Calculate $V(3)$ and find x such that $V(x) = 64$.

Rep. $V(3) = 27$; $x^3 = 64$, $x = 4$.

5. Red thread · Flash [REP] In which quadrant (numbered from I to IV in the trigonometric direction) is found $M(a;b)$ with $a < 0$ and $b > 0$?

Rep. Quadrant II (top left).

Form 88 game. 1st Jul 2027 Annual review and consolidation

1. Warming · Flash [GEO] Hypotenuse of a right triangle with legs 5 and 12?

Rep. $\sqrt{25+144}=13$.

2. Heart · Flash [REP] Midpoint of $[AB]$ with $A(-3;2)$ and $B(5;-4)$?

Rep. $\left(\frac{-3+5}{2};\frac{2-4}{2}\right)=(1;-1)$.

3. Heart + · Write [DEM] Show that $A(0;0)$, $B(2;1)$, $C(-1;2)$ form an isosceles right triangle A.

Rep. $AB^2=5$, $AC^2=5$; $BC^2=9+1=10=AB^2+AC^2$: Reciprocity of Pythagoras

4. Back · Flash [NUM] Give an approximate value of $\frac{1}{3}$ à 10^{-2} Is this number decimal?

Rep. 0.33. No: Decimal writing of $\frac{1}{3}$ is unlimited (0.333 . . .).

5. Red thread · Problem [IVD] What is the smallest integer that gives remains 1 in division by 4 and 6, More than 1?

Rep. $1+\text{PPCM}\,(4\,;\,6)=13$.